2027 EDISON AWARDS
NOMINEE

PUBLIC SAFETY · SECURITY · DIGITAL INTEGRITY

FOR MILLENNIA
CIVILIZATION & ECONOMIC LIFE

Trust — the oldest question of civilization.

How can we verify before we act?



SMVRS™

INDEPENDENT VERIFY

Proof Before Effect.

Portable semantic proof across devices and locations.

SIGNED RECORD
- Identity & Authority
- Evidence
- Events & Time
- Policy & Rules

TWO INDEPENDENT VERIFIERS
- Python Reference
- Node.js Reference
- Same Input Same Rules

EXPLAINABLE RESULT
- Verified / Failed
- Explicit Reason Codes
- Detects Tampering
- Consistent Results

FOR EVERYONE · EVERY FAMILY · EVERY ENTERPRISE · EVERY NATION



INDIVIDUALS
Personal records, identity and rights



FAMILIES
Health, education, assets and services



ENTERPRISES
Contracts, compliance and trusted operations



INSTITUTIONS
Public services, regulation and trusted digital state



NATIONS & REGIONS
Interoperable trust across jurisdictions

Inventor: Dr. FURONG BEI · SMVRS.com · ATMS.com · contact@atms.com

THE QUESTION

Trust — civilization's millennia-old conundrum.

For thousands of years, economic life has depended on records, and people have largely had to rely on whoever held them. Trust is a foundation of civilization's economy. The question has always been the same: how can a receiver verify before acting?

c. 3200 BCE
Clay tablets record grain, debts and ownership.

1494
The first printed account of double-entry bookkeeping spreads checkable ledgers.

TODAY
Databases and digital signatures move records across systems, borders and machines.

SMVRS answer:

Proof Before Effect.

A portable record designed so that any receiver can independently verify it, in any place, any scenario, at any time, before money, care, goods or machines move.

AT A GLANCE

SMVRS™ Independent Verify: portable proof with independent semantic verification.

Beyond the signature, any receiver can check a record’s evidence, events, time and pinned rules. Two separately built verifiers mark consistent records VERIFIED and mismatched ones FAILED, with exact reasons. Re-verified on servers in Oregon and Northern California.

01 · CONCEPT

A digital signature shows that bytes are unchanged. SMVRS adds a second layer: it checks whether a record’s evidence, events, time and rules still agree, without relying on the issuing system.

02 · VALUE

Python and Node.js verifiers, built separately, recompute the proof on the receiver’s side. In controlled tests an intact record passed all 18 checks; altered evidence failed with the exact reason code. Standard cryptography, no special hardware.

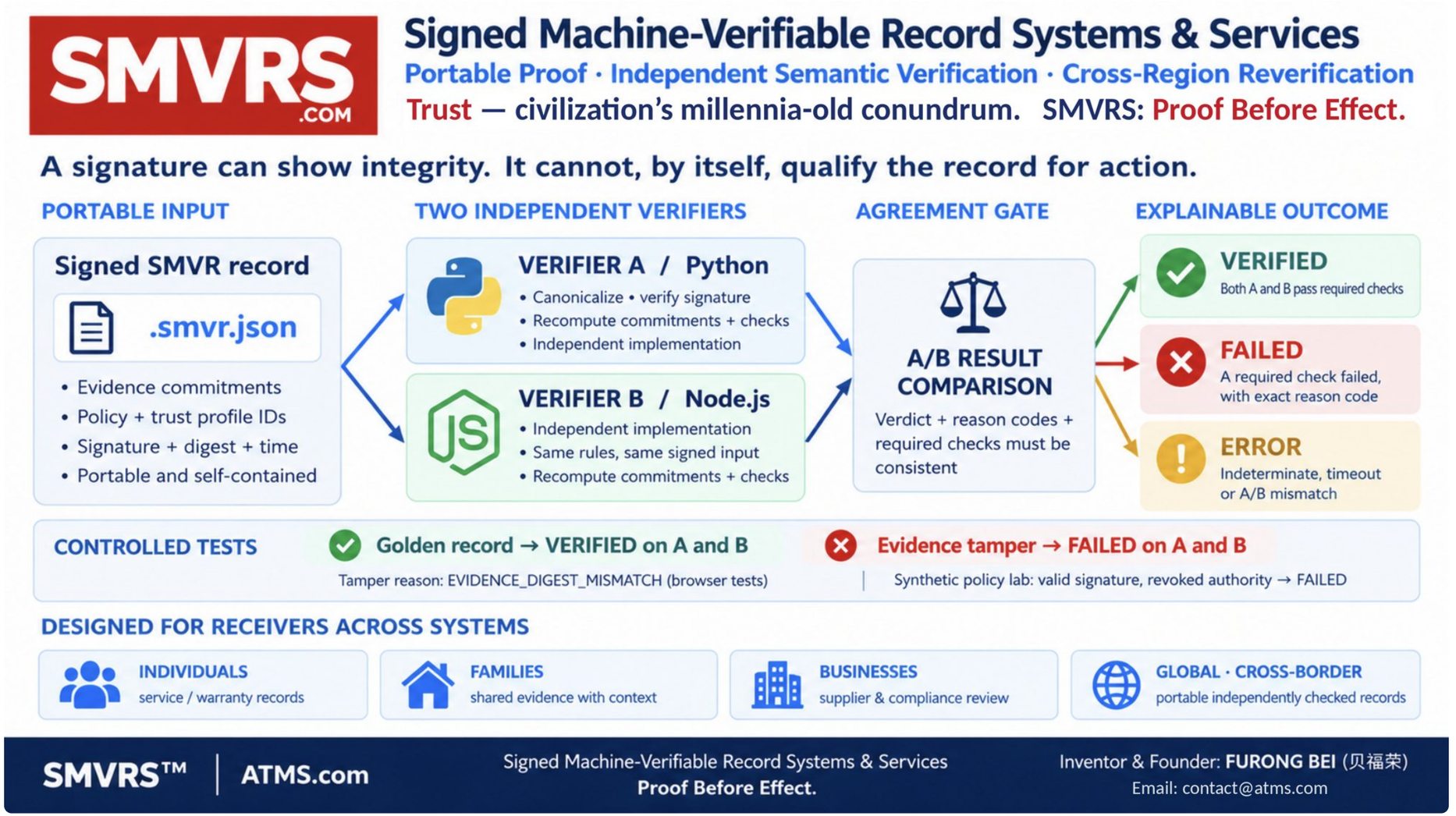
03 · IMPACT

SMVRS is designed so that individuals, families, enterprises, institutions and nations can check a record before acting. As AI agents act at machine speed, proof before effect can become a matter of public safety and digital integrity.

> **Trust is not declared; it is independently verified.**

THE INNOVATION

How SMVRS Independent Verify works



Portable signed record → two independent verifiers (Python A, Node.js B) → A/B agreement gate → explainable outcome: VERIFIED, FAILED with a reason code, or ERROR.

WATCH · READ · CHECK

Every statement in this document can be followed to a source.

60-second film	youtu.be/HiTej_EingU
Edison nominee page	edisonawards.com/nominees/smvrs-independent-verify
Product	smvrs.com
Company	atms.com
Public beta	bei.app
Patent records (Justia)	patents.justia.com/inventor/furong-bei
Patent records (Google)	patents.google.com/?inventor=Furong+Bei
Contact	contact@atms.com

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At a glance · p. 3

The innovation · p. 4

01 Concept · pp. 6–15

02 Value · pp. 16–20

03 Impact · pp. 21–23

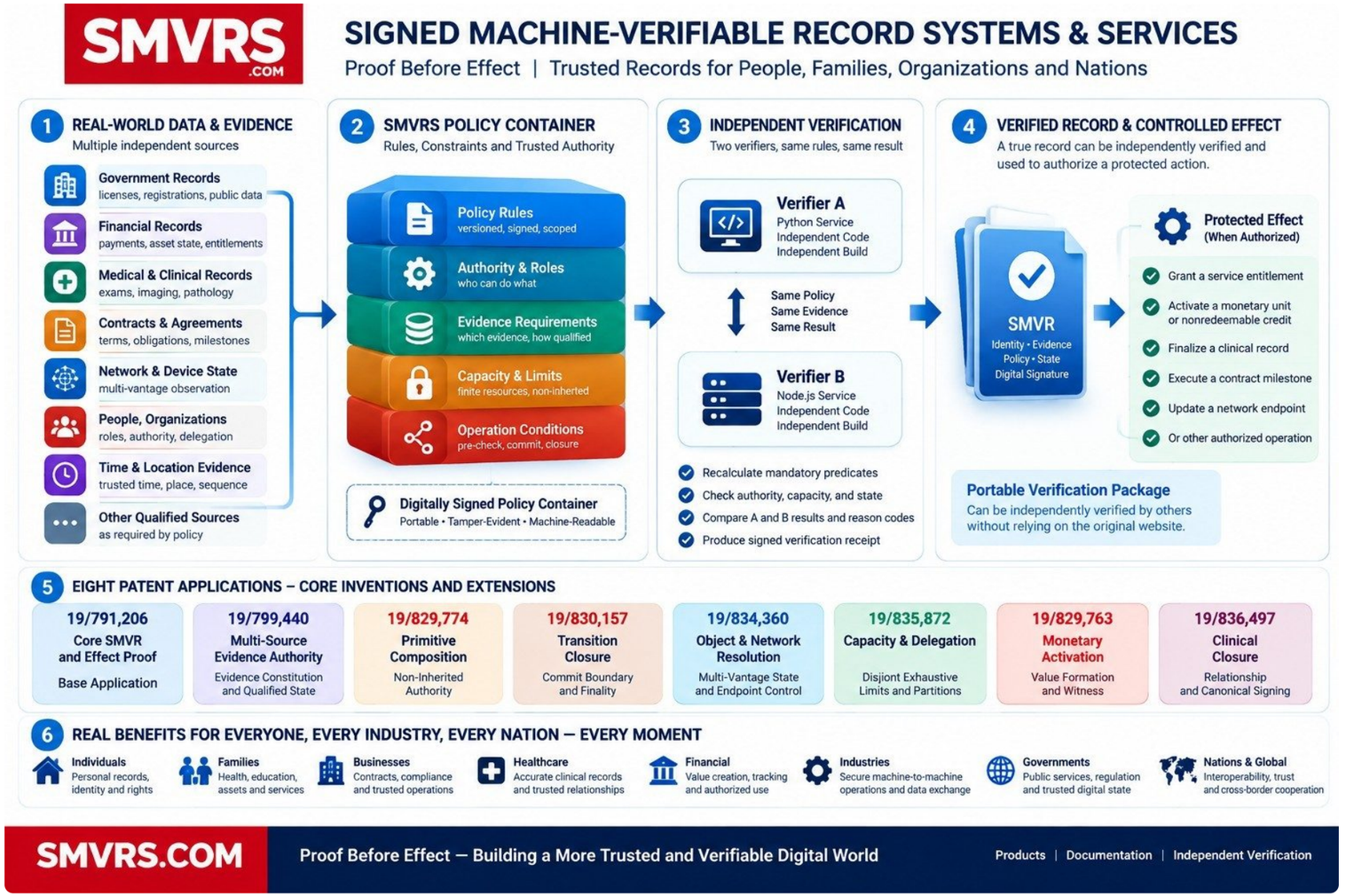
Architecture · pp. 24–28

Patent applications · p. 29

Inventor and contact · p. 30

Protocol BEI-SMVR-0.1 · Verification profile SMVR-VERIFY-1 · Golden record smvr:6d5845d693596ca36e0c0de5

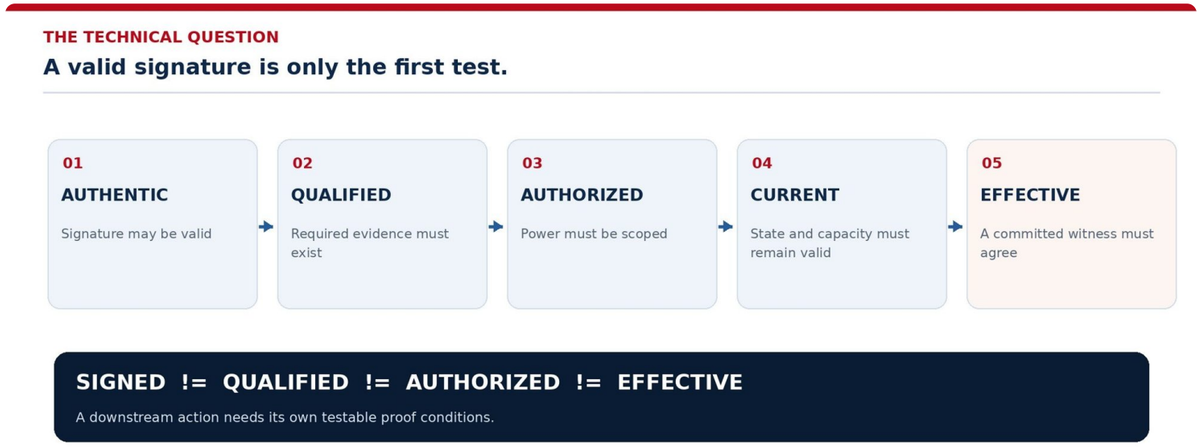
Core invention: from record to qualified action



Qualified real-world inputs, a signed policy, two independent verification paths, a portable record and a protected-effect boundary. The effect stages illustrate the broader architecture described in the patent applications.

A valid signature is the start of trust, not the whole of it.

Authentic bytes can still carry incomplete evidence, outdated authorization, a wrong relationship or a result that never took effect. SMVRS addresses this missing link in cross-system digital trust.



Machine-readable conditions

SMVRS separates cryptographic authenticity, semantic relationship validity, evidence sufficiency, bounded authority and the operation’s actual result.

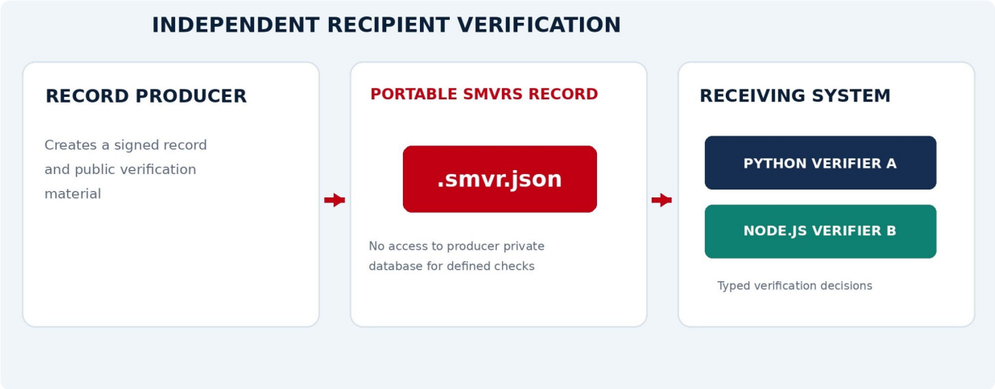
A falsifiable rule

A signed but incomplete or wrongly linked record fails the required check instead of inheriting trust from a logo, a dashboard, a copied source or a majority vote.

From trusting a producer’s claim to independently checking the conditions for relying on it.

A portable record can be checked away from the issuing system.

The SMVR record binds actor, behavior, evidence, policy, time and claimed decision. Independent Python and Node.js implementations recompute the validation rules and report typed failures.



Demonstrated in controlled tests

Golden-record acceptance; rejection of behavior and evidence tampering; semantic-binding cases; agreement between two independent verifiers.

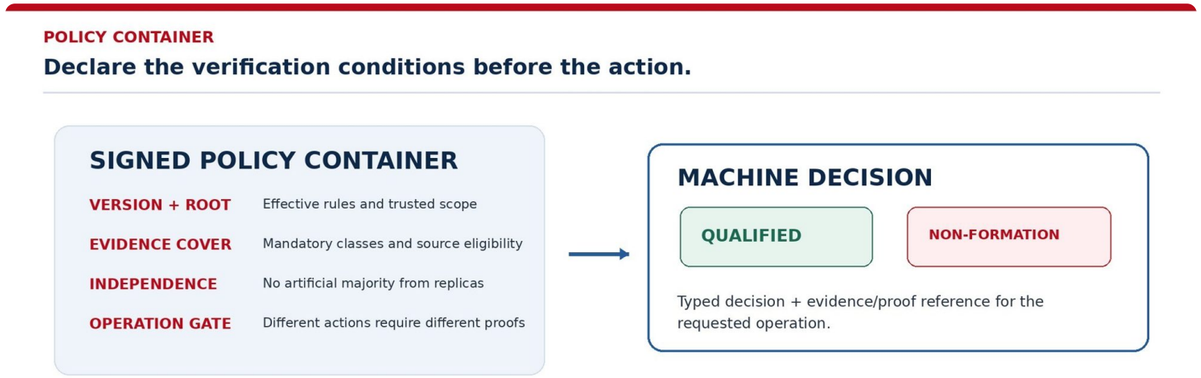
Beyond the signature

A signature confirms the bytes. SMVRS also checks evidence, authority, time and rules, so a receiver sees which stated conditions a record satisfies.

Independent verification is demonstrated today; effect qualification extends the same principle to the moment of action.

The applicable rule is pinned before the decision.

In the SMVRS architecture, a signed Policy Container is designed to bind context, effective version, trust source, mandatory evidence categories, scope, operation and revocation. A proposed machine action qualifies only when the applicable predicates are satisfied.



Why it matters

Without versioned rules, two services may verify the same bytes yet apply different authority, purpose, time or privacy requirements.

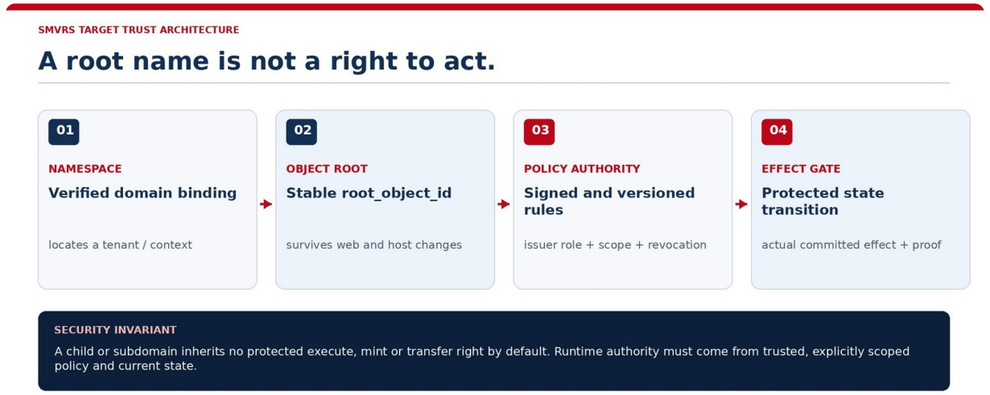
Prototype and roadmap

A signed evidence-policy experiment runs on synthetic data. A cross-service Policy Container registry and a non-bypass action gate are the next extensions.

Policy becomes machine-testable qualification, applied the same way by every receiver.

A web address can locate a record. Only trusted policy can authorize an action.

SMVRS distinguishes a verified website or tenant name from persistent object identity, an approved issuer and explicitly bounded permissions. A child object is designed to start with no protected execution right unless a current signed Policy Container grants one.



Stable identity beyond a URL

A root object stays linked to its proofs even if its displayed host or domain changes.

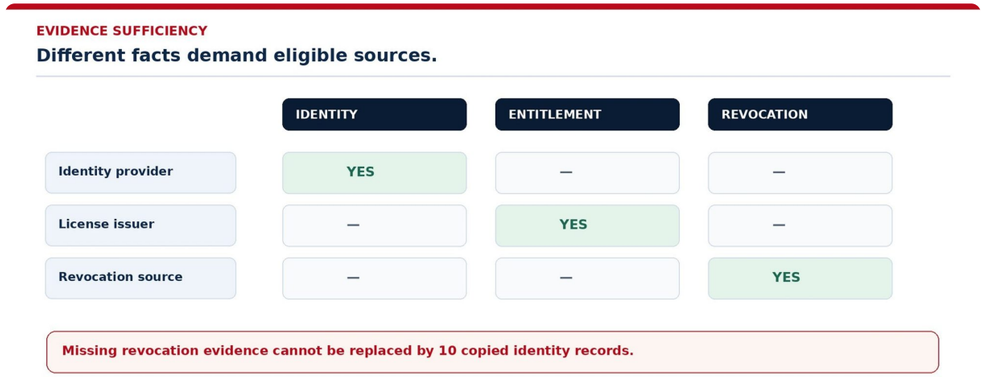
Fail-closed delegated rights

A parent signature or an inherited name cannot silently confer mint, transfer or execution privileges.

> **Names aid discovery; machine authority comes from authenticated trust, scoped grants and current-state checks.**

Each required kind of evidence must come from an eligible source.

An operation can require evidence of identity, entitlement, revocation and observed effect, each from its own eligible authority. Correlated copies do not count as independent corroboration.



A sharper question

Instead of asking which database is most trusted, SMVRS asks whether the evidence classes this operation needs are complete and independently eligible.

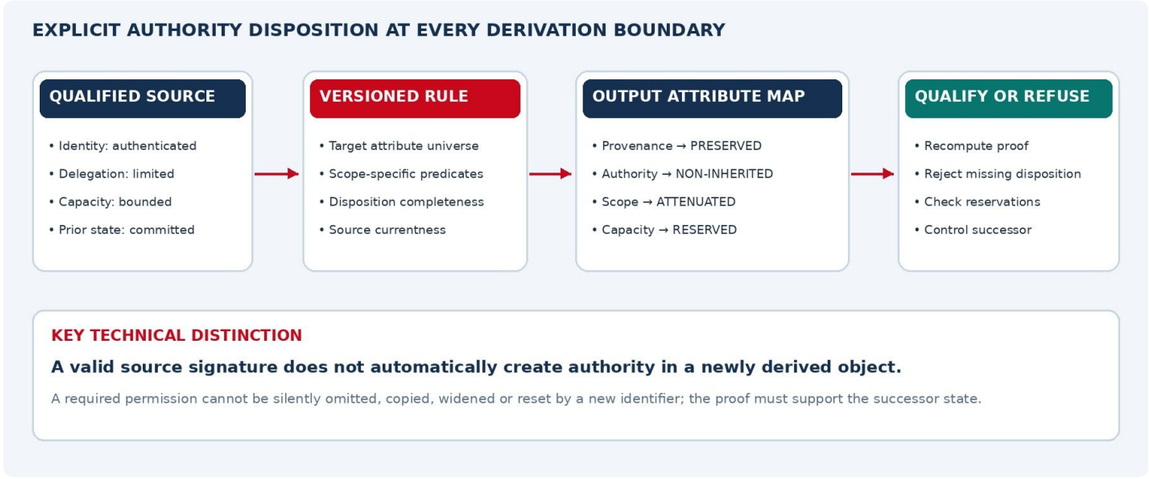
Tested on synthetic fixtures

Signed test fixtures accept a complete qualifying set and reject a missing mandatory class, an out-of-scope assertion or disallowed correlated origins.

No single institution needs to hold a universal master record.

Provenance can persist. Permission is qualified again.

Primitive composition separates where a derived object came from and what protected authority it may exercise. Attribute disposition, allowed scope and live predecessor conditions constrain each successor state.



Scope-limited derivation

Derivation rules are designed so that a child object, delegated right or transformed asset cannot silently gain new purpose, quantity, duration or privileged operation through inheritance.

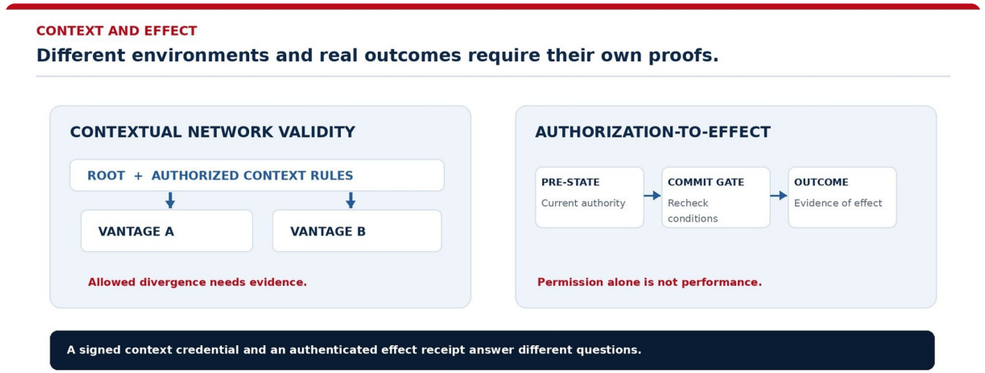
At the commit boundary

Transition closure rechecks current state, policy, replay and authorized effect when an operation is committed, not only when it is proposed.

> **Derived objects keep their history but must earn their authority.**

Correct observations can differ. A claimed effect must be shown.

Rooted multi-vantage resolution qualifies differences between observations under an authenticated policy. The foundational application addresses the match between authorized preconditions and demonstrated state effects.



Contextual verification

A network difference can be permissible under policy; a failing mandatory observation cannot be outvoted by a generic majority.

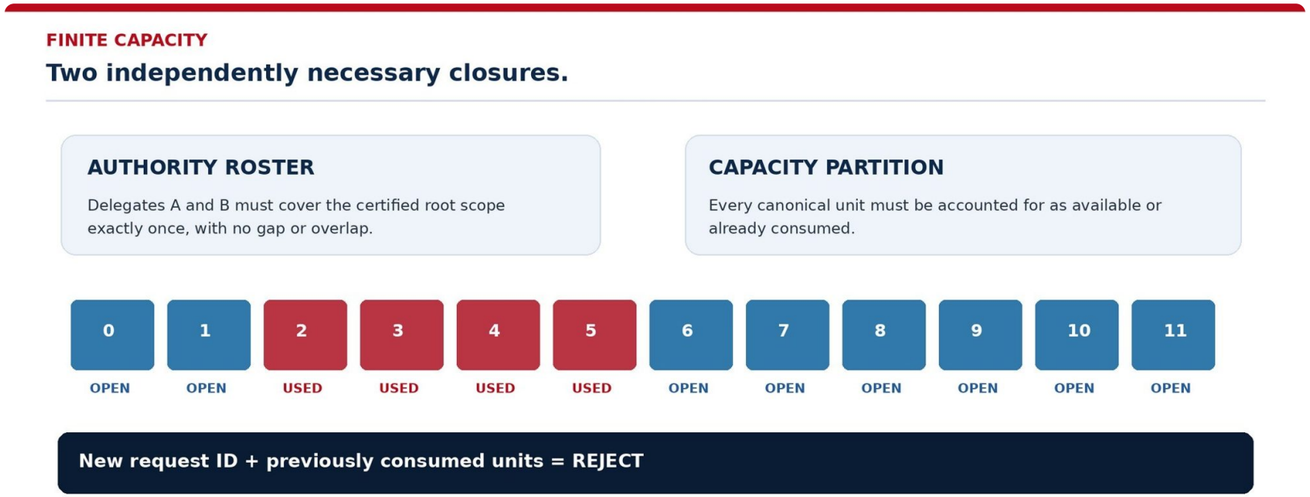
Effect correspondence

An effect receipt is designed to bind authorized actor, policy, predecessor state, effect evidence and successor state before a recipient accepts that an operation occurred.

Context is qualified by policy, and effects are confirmed by evidence.

An authorized delegate must not spend the same unit twice.

Certified capacity requires two complete checks: who is entitled to allocate units, and which exact units remain available. Both must hold for a serialized, witnessed successor state.



Laboratory capability

Test-key signatures, a complete authorized roster, finite-unit accounting, a local serialization point and detachable historical witness checks.

Anti-regeneration goal

Alias changes, split objects or new request identifiers must not turn an already consumed unit into newly available capacity.

Entitlement and availability are checked together before a unit is allocated.

The same rules, tested in two demanding domains.

Research adapters apply SMVRS principles to bounded test-value activation and to typed clinical-record relationships, each as a separate technical extension.



Bounded test-value activation

A simulated monetary controller checks pinned policy and version, source-use correlation, commit-time witness and historical validity, using simulated values only.

Clinical relationship closure

Synthetic patient, specimen and result graphs expose incorrect signed links and distinguish a candidate from a committed record, supporting data integrity rather than diagnosis.

Narrow, repeatable machine checks that each sector can adopt on its own terms.

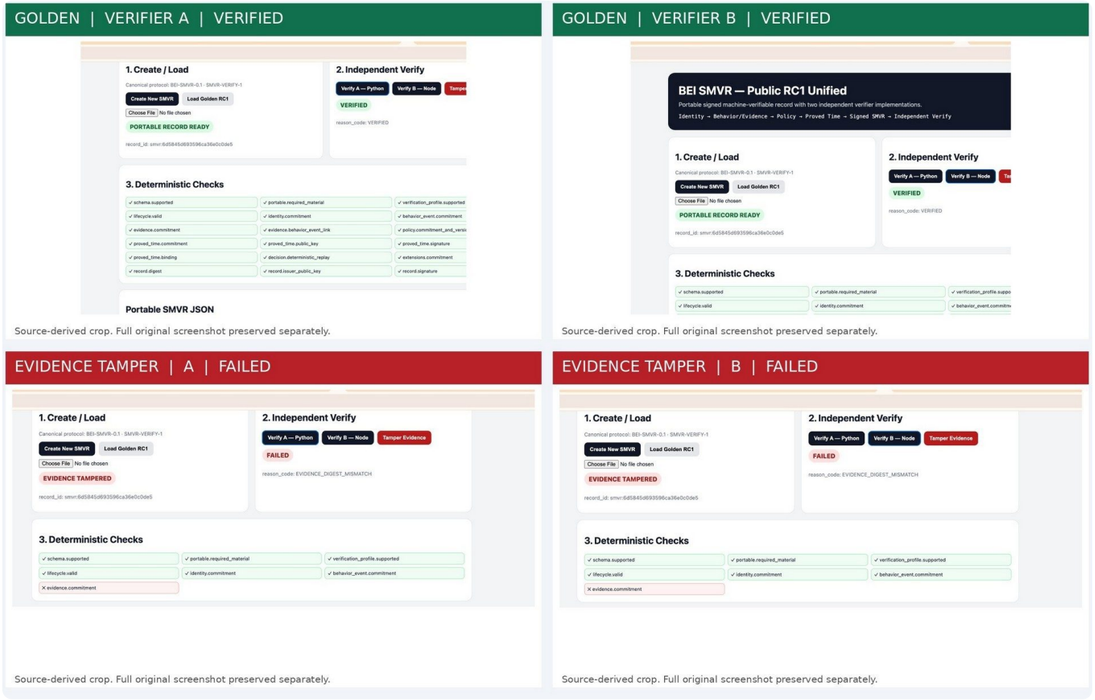
Same record, two verifiers, the same answer.

INPUT	VERIFIER A · PYTHON	VERIFIER B · NODE.JS
Synthetic signed Golden record	VERIFIED · 18 checks	VERIFIED · 18 checks
Synthetic evidence tamper	FAILED	FAILED

Reason code on both verifiers: `EVIDENCE_DIGEST_MISMATCH` at `evidence.commitment`.

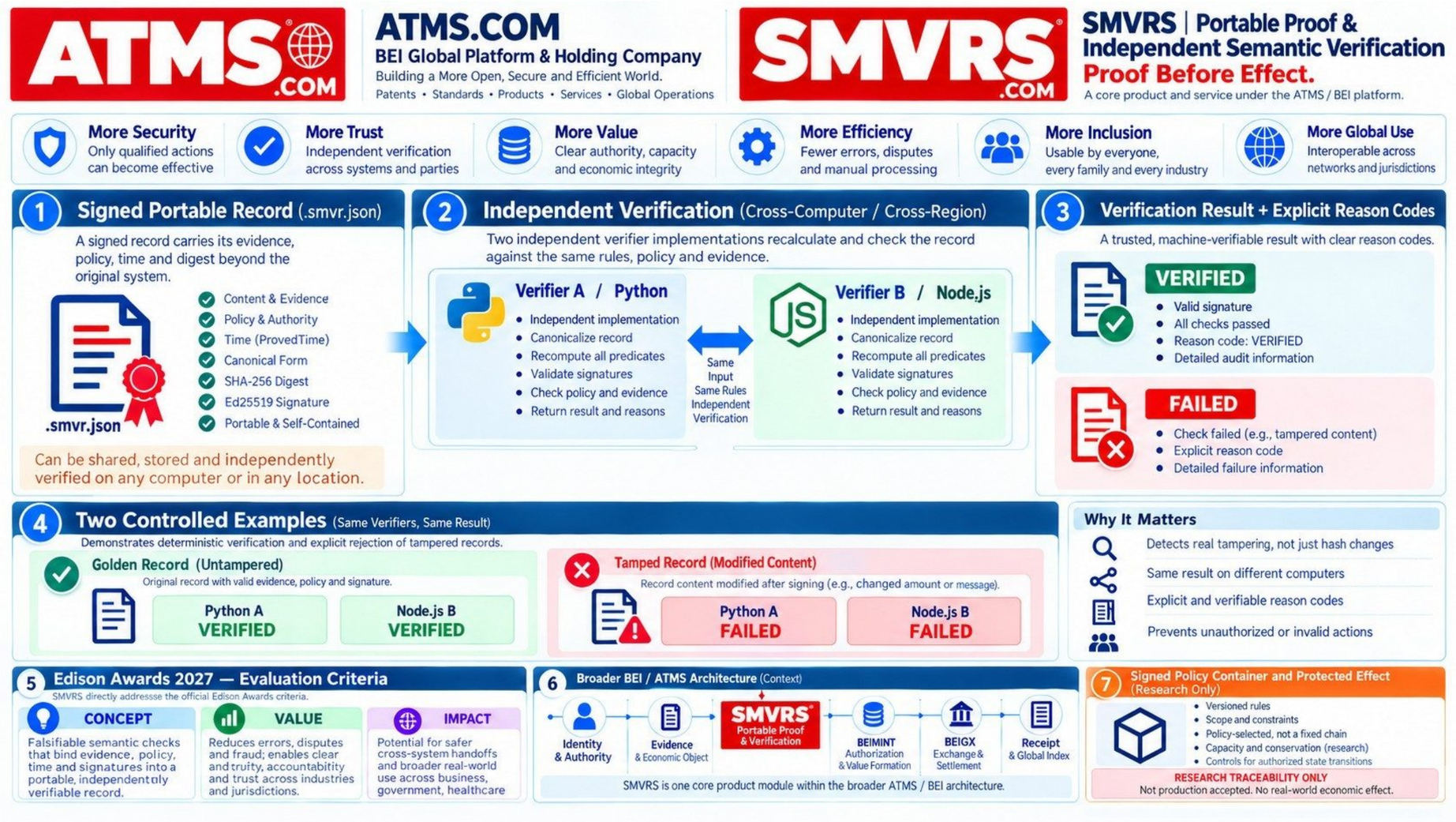
Record: `smvr:6d5845d693596ca36e0c0de5`

Cross-region: in the inventor's engineering tests, a record created on a server in Oregon was re-verified by both verifiers on a server in Northern California; a tampered copy was rejected.



Controlled browser tests on synthetic signed records, October 2026.

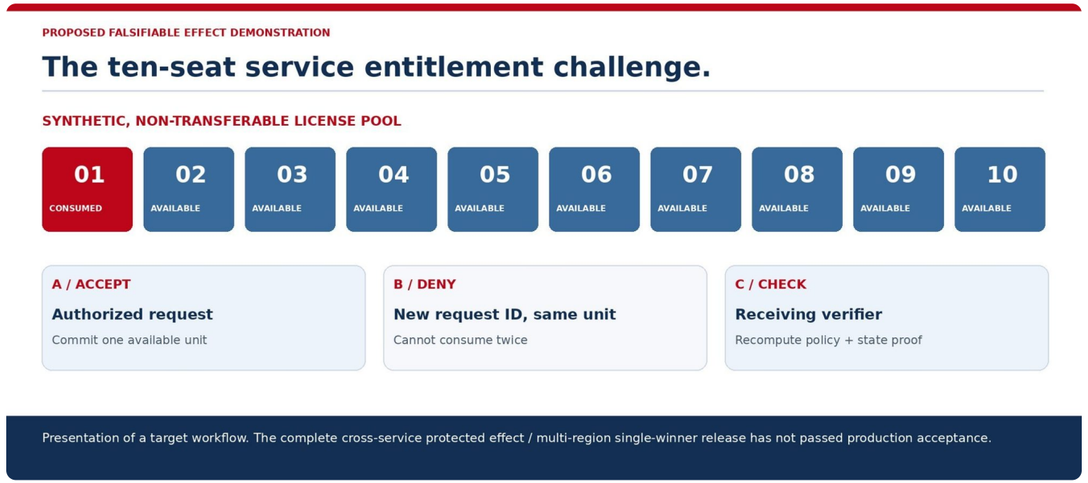
Four controlled cases mapped to the Edison criteria



Golden record VERIFIED on A and B; evidence tamper FAILED on A and B, each with its exact reason code.

A clear first customer test: software-license allocation.

An Economic Node with ten non-redeemable service seats offers a falsifiable first use case. A requester may hold a valid credential yet lack entitlement, revocation coverage or unused licensed capacity.



Useful failure cases

Rejects missing entitlement evidence, an outdated right, correlated pseudo-confirmations, an unauthorized delegate or repeat use of the same seats.

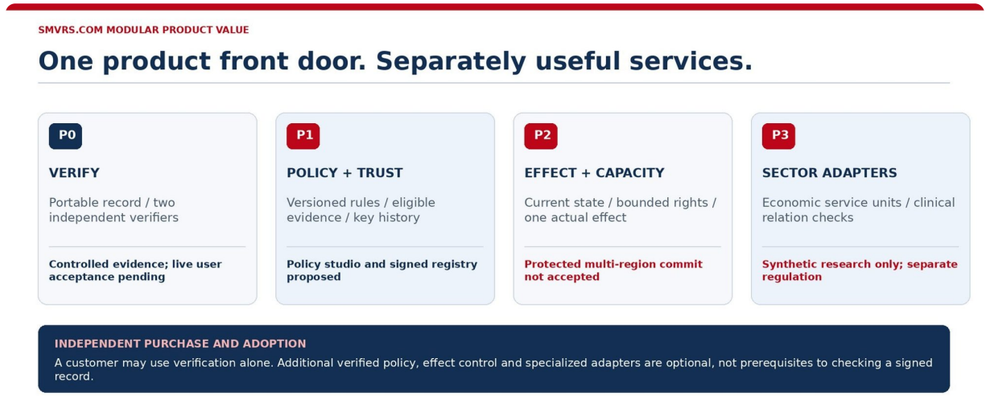
Practical package

Portable record verification is offered first; evidence and finite-capacity controls are designed as added modules toward an end-to-end customer workflow.

> **Value is measured by the decisions a system can independently justify.**

Independent verification first. Deeper assurance when an operation needs it.

SMVRS.com is one product entrance with independently useful editions: portable Verify, signed Policy and eligible Evidence, protected Effect, and sector adapters.



Near-term customer outcome

A recipient uploads a signed record, sees two independently computed results and exports the comparison.

Longer-term customer outcome

Organizations add signed policy qualification or scoped effect control as their security, operational and sector requirements call for it.

Each edition is useful on its own; none requires adopting the monetary or clinical modules.

Evidence is stronger when failure is testable.

2

independent verifier implementations

Python and Node.js, built separately from one specification.

13

archived negative vector classes

Each kind of failure is preserved as a repeatable test.

79/79

local synthetic laboratory tests passed

Policy, evidence and capacity fixtures, run on synthetic data.

Cost

Standard cryptography (SHA-256, Ed25519), no special hardware, designed for integration through APIs with software, networks, machines and IoT devices.

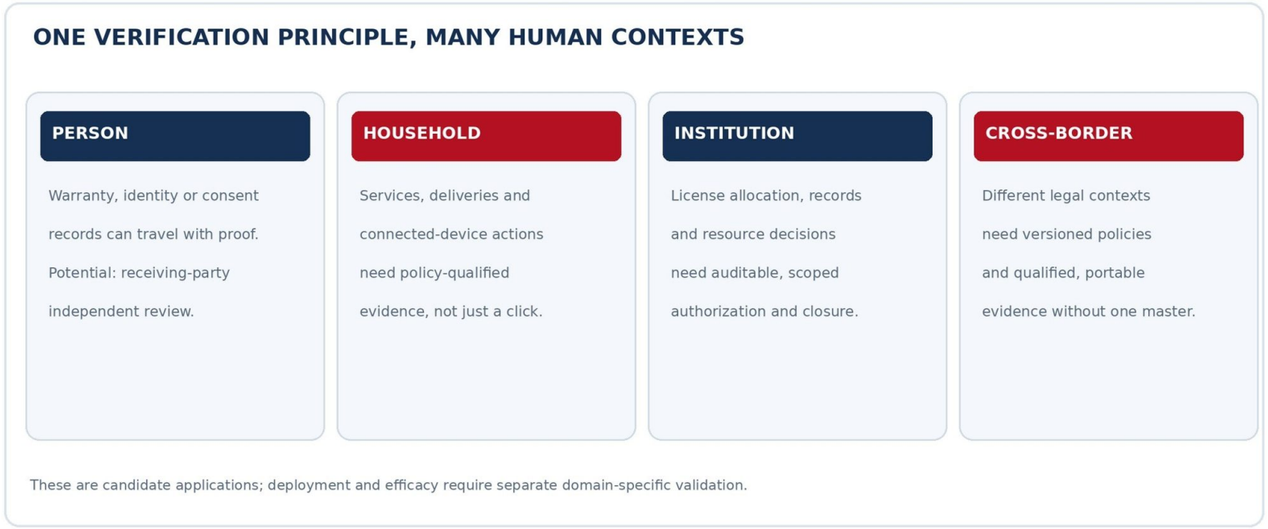
Benefit

Designed to detect an inconsistent record before funds, care or goods are released; in testing, each refusal names the check that failed.

Low adoption cost, high-consequence protection.

Built for people, families, institutions and nations.

A recipient checks a digital assertion against its applicable rules, rather than depending on the producer’s success label alone.



Everyday scale

A person, household or small business can receive a portable result and verify who asserted it, which evidence and policy apply, and whether the record is current.

Across systems and borders

Institutions and regional services exchange proofs without assuming a single database or universal authority, each applying its own law, privacy and standards.

Any place, any scenario, any time: trust that travels with the record.

A layered service each industry can adopt at its own pace.

SMVRS supports independent verification as a stand-alone service, with optional evidence qualification, policy, effect and sector components where deeper controls are needed.



A shared architectural direction; today, only the verifier and separate local research components have controlled tests.

Immediate service

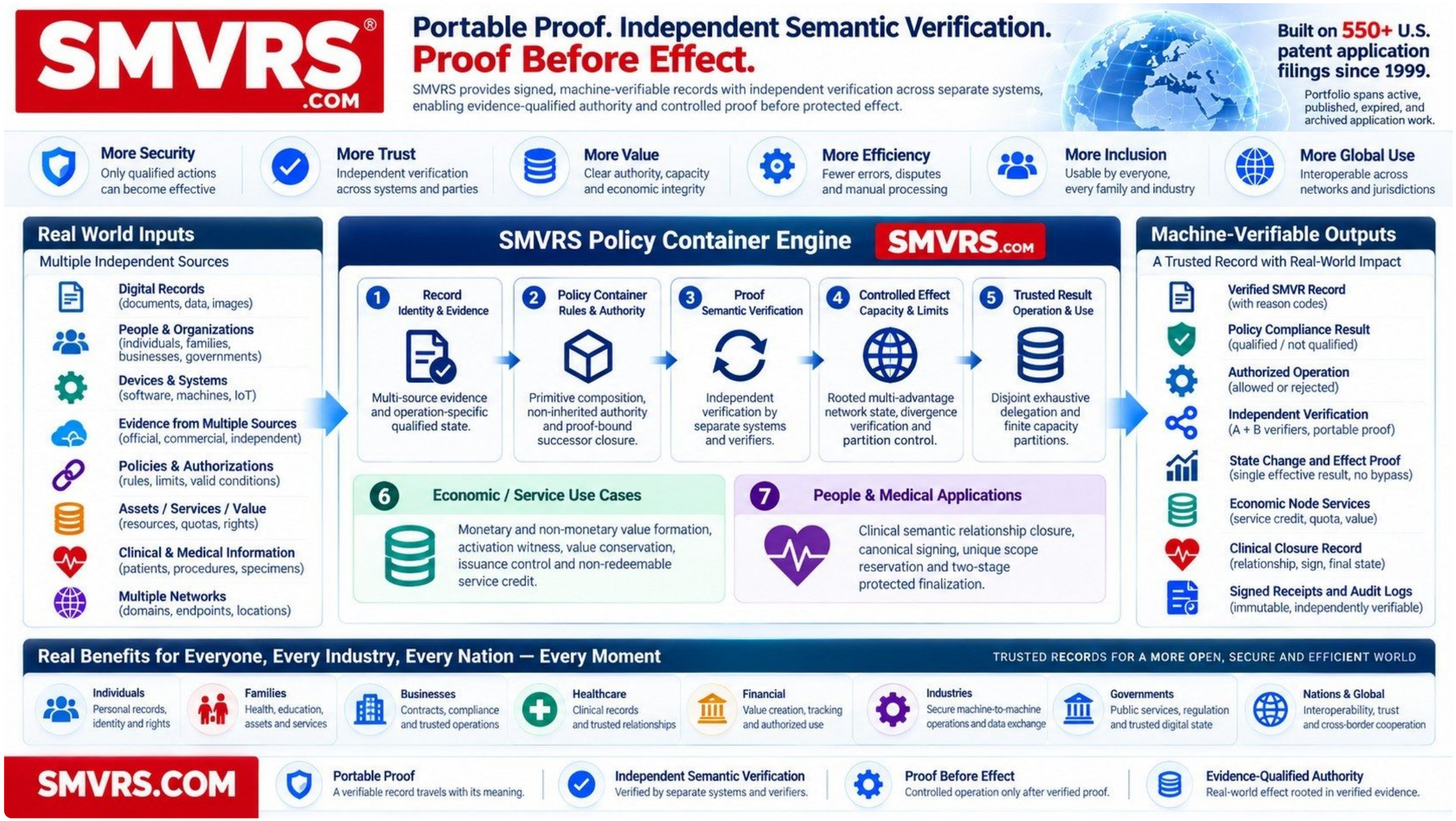
Record receipt, detached check, explicit result and reason code, and evidence export for institutional handoffs.

Longer-term services

Scoped policy decisions, capacity certificates, signed effect receipts and adapters for software, enterprise records, devices and regulated systems.

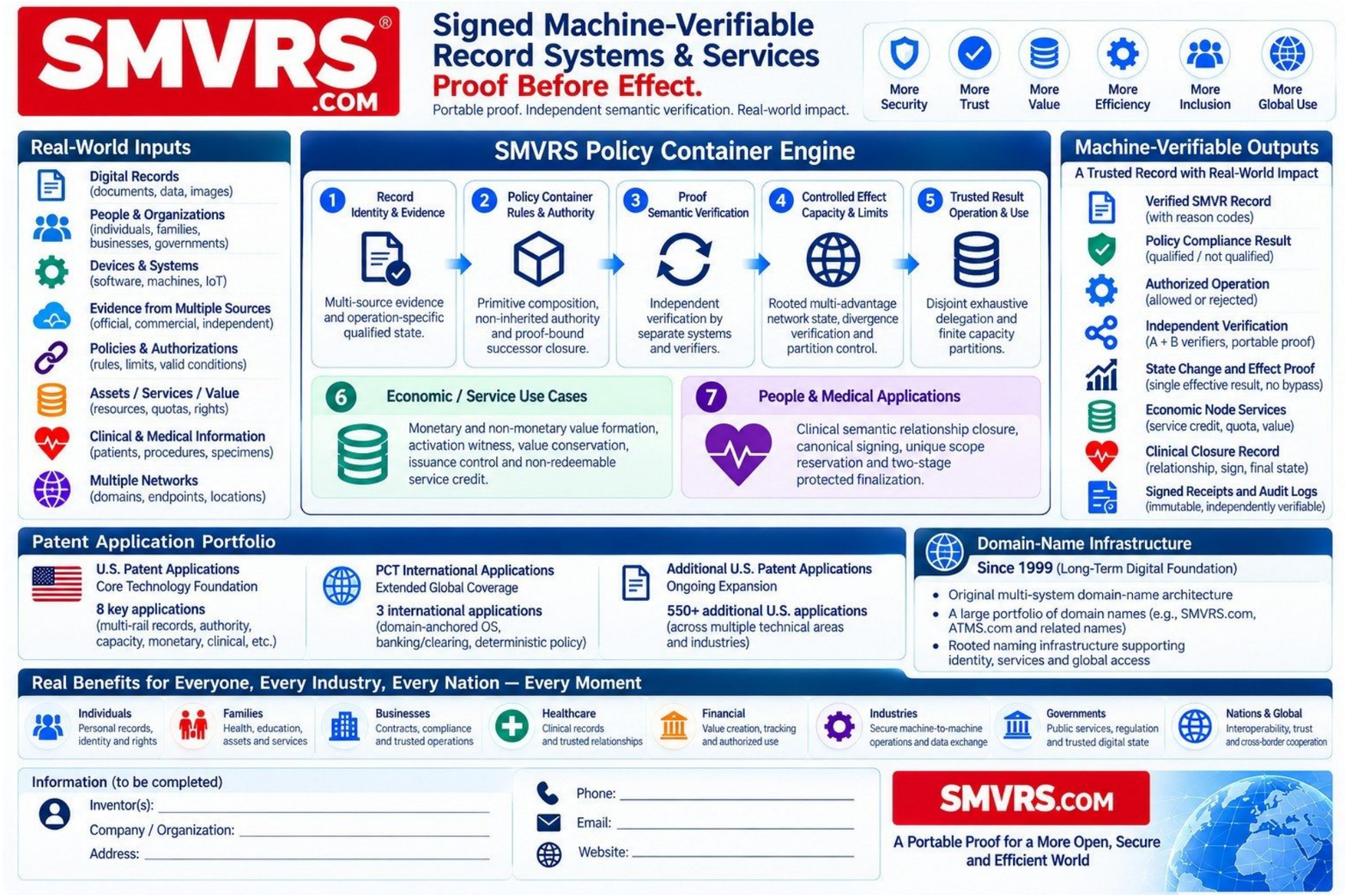
As AI agents act at machine speed, proof before effect is designed to serve as a public-safety layer.

Proof Before Effect: reach and long-term roots



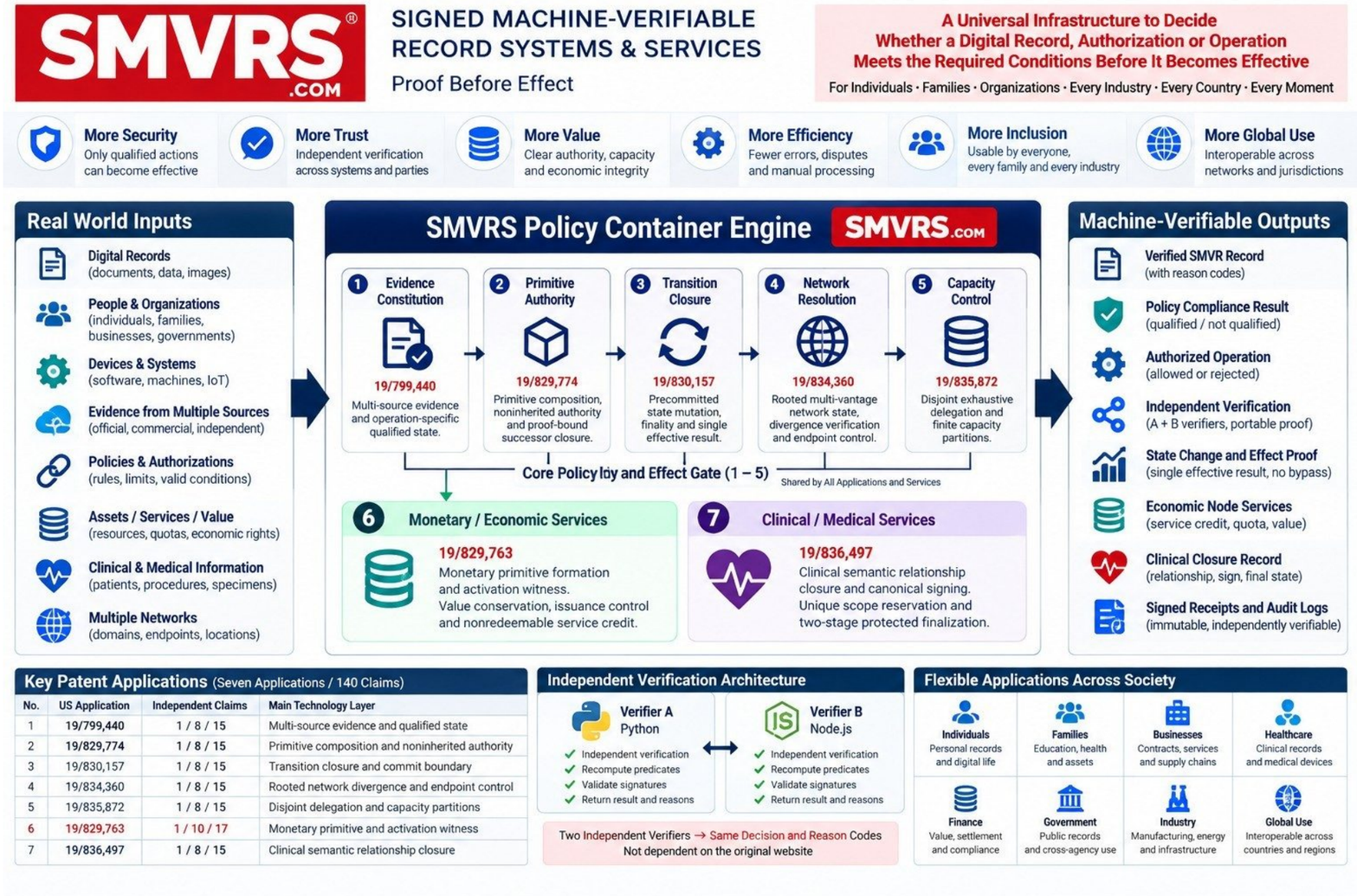
Portable proof, policy control and public value for every person, family, industry and nation. 1999 marks the start of the inventor's domain and digital-infrastructure program.

Complete product, patent and domain landscape



Policy Container Engine, economic and clinical extensions, patent application portfolio, rooted-domain assets and application areas.

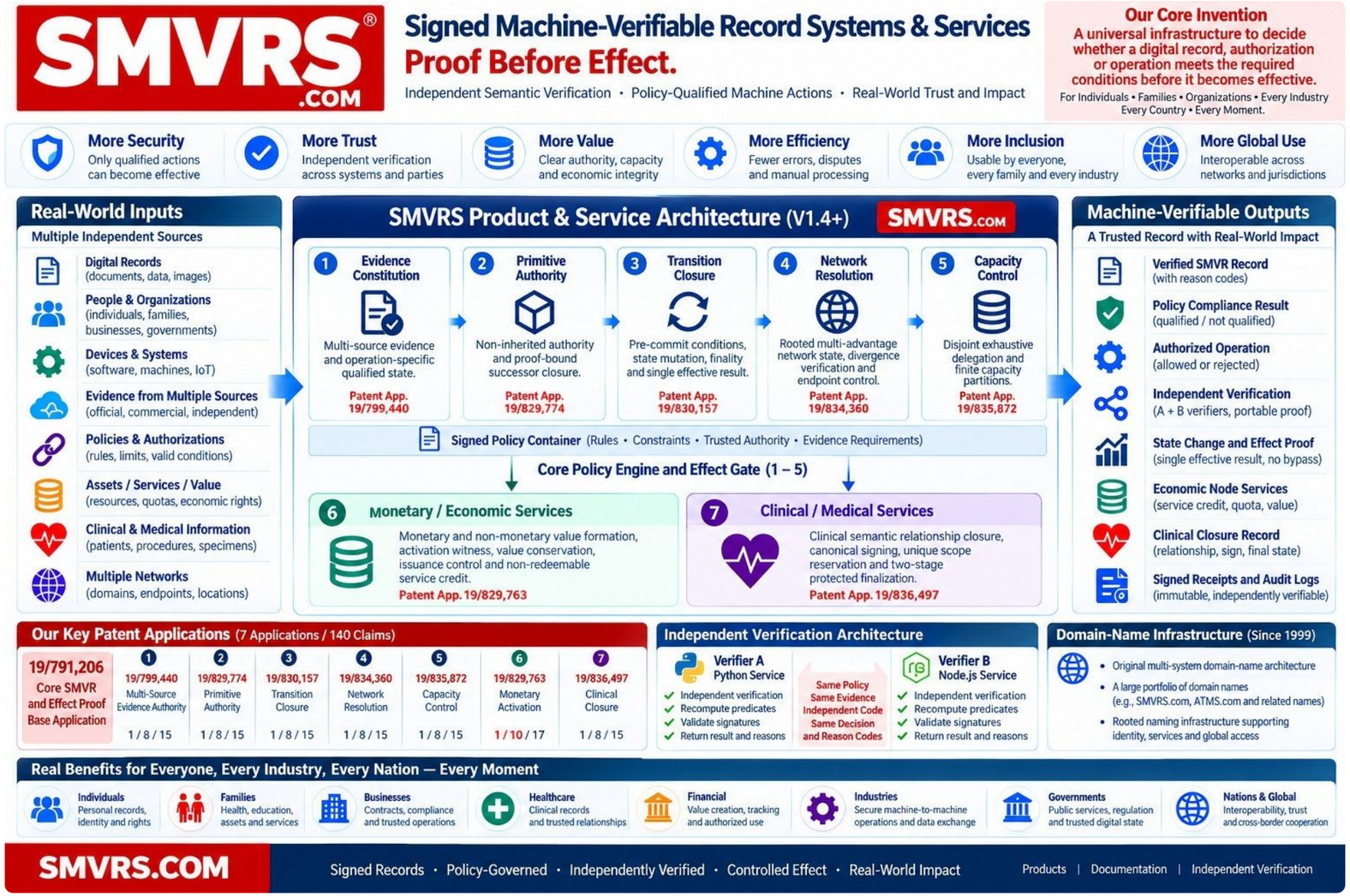
Seven technical layers and independent verification



Seven technical application layers with separate A/B verifier roles and industry inputs and outputs. Application numbers identify filings.

ARCHITECTURE · FIGURE 5

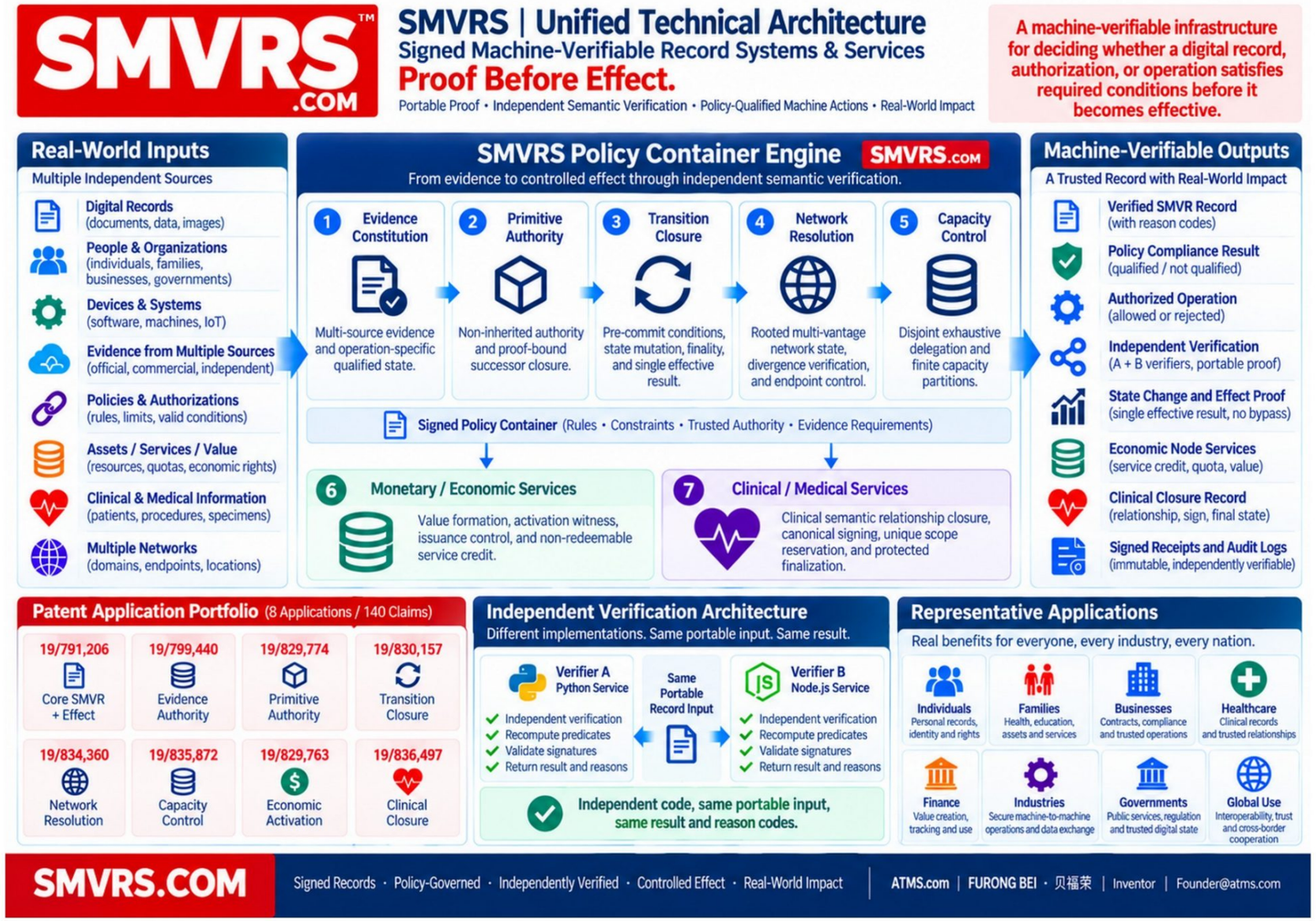
Detailed patent-core and industry blueprint



Evidence, primitive composition, transition closure, rooted network resolution, capacity, and economic and clinical specializations.

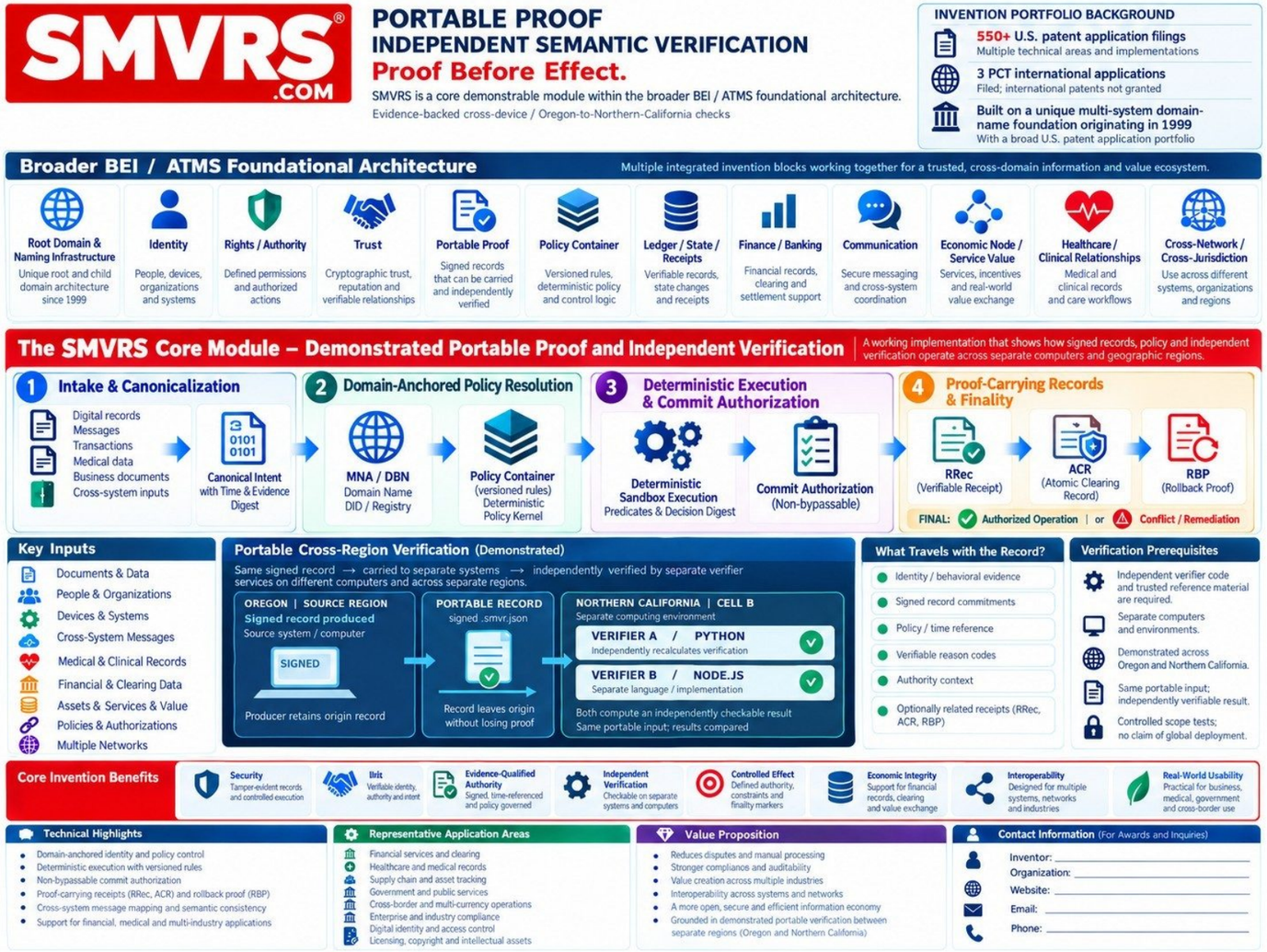
ARCHITECTURE · FIGURE 6

Eight related applications, one controlled system



Signed policy, five technical layers, economic and clinical extensions, and dual verifiers.

BEI × ATMS foundation, portable receipts and future extensions



Inventor research illustration of rooted-domain identities and portable receipt concepts within the broader BEI × ATMS architecture.

PATENT APPLICATIONS

Eight related U.S. applications, distinct technical roles.

19/791,206	Authorization–Effect Correspondence
19/799,440	Evidence-Scoped State Formation
19/829,774	Primitive Composition / Non-Inherited Authority
19/830,157	Transition Closure / Commit Boundary
19/834,360	Multi-Vantage Contextual Resolution
19/835,872	Certified Capacity / Dual Closure
19/829,763	Monetary Control / Activation Proof
19/836,497	Clinical Semantic Relationship Closure

556

U.S. patent applications filed naming the inventor

Across the broader BEI × ATMS portfolio, of which the eight listed here are directed to SMVRS methods.

Since 1999

A domain-rooted naming and digital-infrastructure initiative, developed by the inventor and self-funded.

Public records: patents.justia.com/inventor/furong-bei · patents.google.com/?inventor=Furong+Bei

2027 EDISON AWARDS

NOMINEE

SIGNED MACHINE-VERIFIABLE RECORD SYSTEMS & SERVICES



SMVRS™

INDEPENDENT VERIFY

Proof Before Effect.

TRUST — CIVILIZATION'S
MILLENNIA-OLD CONUNDRUM

How can we independently verify before we act?

FOR EVERY PERSON | EVERY FAMILY | EVERY NATION | EVERY REGION

Toward a verifiable foundation for civilization and economic life.

Inventor: Dr. FURONG BEI

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